Compounds of Transitional elements

141. $4K_2Cr_2O_7 \xrightarrow{\text{heat}} 4K_2CrO_4 + 3O_2 + X$. In the above reaction X is
(a) CrO_3 (b) Cr_2O_7
(c) Cr_2O_3 (d) CrO_5

142. Monds process is used for
(a) *Ni* (b) *Al*
(c) *Fe* (d) *Cu*

143. Stainless steel is an alloy of
(a) Copper
(b) Nickel and chromium
(c) Manganese
(d) Zinc

144. Percentage of silver in German silver is
(a) 0%
(b) 1%
(c) 5%
(d) None of these

145. Which process of purification is represented by the following scheme $\underset{\text{impure}}{Ti} + 2I_2 \xrightarrow{250^oC} TiI_4 \xrightarrow{1400^oC} \underset{\text{pure}}{Ti} + I_2$
(a) Cupellation
(b) Poling
(c) Electrolytic refining
(d) Zone refining
(e) Van-Arkel process

146. Which of the following sulphides when heated strongly in air gives the corresponding metal
(a) Cu_2S (b) CuS
(c) Fe_2S_3 (d) *FeS*
(e) *HgS*

147. Guignet's green is known as
(a) $Cr_2O_3.\ 2H_2O$
(b) $FeO_3.2H_2O$
(c) Cu_2O_3
(d) $FeCO_3.Cr_2O_3$
(e) $FeO\ .\ Cr_2O_3$

148. Vanadium (III) oxide is a strong
(a) Drying agent
(b) Oxidising agent
(c) Reducing agent
(d) Wetting agent
(e) Precipitating agent

149. Stainless steel does not rust because
(a) Chromium and nickel combine with iron
(b) Chromium forms an oxide layer and protects iron from rusting
(c) Nickel present in it, does not rust
(d) Iron forms a hard chemical compound with chromium present in it.

150. The main product obtained when a solution of sodium carbonate reacts with mercuric chloride is
(a) $Hg(OH)_2$
(b) $HgCO_3.HgO$
(c) $HgCO_3$

(d) $HgCO_3.Hg(OH)_2$

151. Which of the following has diamagnetic character

(a) $[NiCl_4]^{2-}$

(b) $[CoF_6]^{3-}$

(c) $[Fe(H_2O)_6]^{2+}$

(d) $[Ni(CN)_4]^{2-}$

152. The solubility of silver bromide in hypo solution due to the formation of

(a) $[Ag(S_2O_3)]^{-3}$ (b) Ag_2SO_3

(c) $[Ag(S_2O_3)]^{-}$ (d) $Ag_2S_2O_3$

153. Brass is an alloy of

(a) *Zn* and *Sn* (b) *Zn* and *Cu*

(c) *Cu*, *Zn* and *Sn* (d) *Cu* and *Sn*

154. Iodine is formed when *KI* reacts with a solution of

(a) $CuSO_4$ (b) $(NH_4)_2SO_4$

(c) $ZnSO_4$ (d) $FeSO_4$

155. Rust is

(a) $FeO + Fe(OH)_2$

(b) Fe_2O_3

(c) $Fe_2O_3 + Fe(OH)_2$

(d) Fe_2O_3 and $Fe(OH)_3$

156. $[Sc(H_2O)_6]^{3+}$ ion is

(a) Colourless and diamagnetic

(b) Coloured and octahedral

(c) Colourless and paramagnetic

(d) Coloured and paramagnetic

157. Which of the following is called white vitriol

(a) $ZnCl_2$

(b) $MgSO_4.7H_2O$

(c) $ZnSO_4.7H_2O$ (d) $Al_2(SO_4)_3$

158. $FeSO_4.7H_2O$ shows isomorphism

(a) $ZnSO_4.7H_2O$

(b) $MnSO_4.4H_2O$

(c) $CaSO_4.5H_2O$

(d) $CaCl_2.2H_2O$

159. Which pair of compound is expected to show similar colour in aqueous medium

(a) $FeCl_2$ and $CuCl_2$

(b) $VOCl_2$and $CuCl_2$

(c) $VOCl_2$ and $FeCl_2$

(d) $FeCl_2$ and $MnCl_2$

160. Which of the following dissolves in hot conc. *NaOH* solution

(a) *Fe* (b) *Zn*

(c) *Sn* (d) *Ag*